

Santiago Lopez Carreon

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<https://github.com/SantiagoELC-Git>

SUMMARY OF QUALIFICATIONS

- Strong interpersonal skills developed through extracurriculars.
- Strong oral and written communication developed through coursework and extracurricular activities.
- Strong computer literacy with general software such as Office 360 and Jupyter developed through coursework, personal interest, and extracurriculars.
- Proficiency in Python developed through work experience, extracurriculars, and coursework. Proficiency with HTML, CSS, and JavaScript developed through work experience and personal projects.
- Proficiency in MATLAB through online course MATLAB on-ramp, with extra practice with MATLAB community problems.
- Lab skills developed through coursework

WORK EXPERIENCE

Data Scientist/Machine Learning Co-op

Jan. 2026 – April. 2026

BBD Insights

Toronto, ON

- Contributing to backend development of risk model governance software, supporting risk model inventory and oversight workflows.
- Developed a functional REST backend API with Python using Flask library
- Developed production-ready systems for AI-driven document summarization using RAG and function-calling patterns with Python libraries (LangChain, google-genai), integrated via Flask; implemented exponential backoff and retry logic for robust API interactions.

LAB EXPERIENCE

CHEM 120L - General Chemistry Laboratory 1 Course

Sept. 2024 – Dec. 2024

Faculty of Science

Waterloo, ON

CHEM 123L - General Chemistry Laboratory 1 Course

Jan. 2025 – April. 2025

Faculty of Science

Waterloo, ON

PHYS 260A - Intermediate Physics Laboratory 1

Sept. 2025 – Dec. 2025

Faculty of Science

Waterloo, ON

VOLUNTEER EXPERIENCE

Wind Tunnel Project

May. 2026 - Present

Faculty of Engineering

Waterloo, ON

- Collected research on polycarbonate welding.
- Helped with the online research collection of electronic components for the wind tunnel construction.
- Responsible for the coding and setting up of the electronics for the wind tunnel, with the VSC, air pressure sensor, and microcontroller

Vice-President of Finance

Jan. 2025 – Aug. 2025

University of Waterloo Undergraduate Physics Society

Waterloo, ON

- Oversaw financial operations of the undergraduate Physics Society, including budgeting, reimbursements, expense tracking, and collaboration with executives and governing boards to ensure transparent and strategic use of funds.
- Managed and reimbursed over \$3,500 in budgets for community events and office supplies, ensuring accurate financial tracking and compliance.
- Utilized spreadsheets and Google Forms to monitor expenses and streamline reporting for teams and the Science Society Board of Directors, which led to clear and fast retrieval of information and a decrease in mistakes from all executives.

- Collaborated with executives to secure funding for events that enhanced student engagement and ensured financial returns benefited the student community.

First Year Representative

Sep. 2024 – Dec. 2024

University of Waterloo Undergraduate Physics Society

Waterloo, ON

- Advocated for first-year Physics students by bridging communication with Society executives and promoting engagement in community events.
- Coordinated timetables and updates between students and executives.

Physics and Math Tutor

May. 2025 – Dec. 2025

University of Waterloo Physics Tutorial Centre

Waterloo, ON

- Delivered one-on-one help in core Mathematics and Physics concepts to engineering first years by taking advantage of metaphors and self-made diagrams to explain concepts
- Applied interpersonal skills to create a comfortable and engaging learning environment through friendly conversation and positive encouragement throughout the session

Volunteer Research Assistant

Nov. 2025 – April. 2026

University of Waterloo Theoretical Chemistry Department

Waterloo, ON

- Developed a testing process with libraries in Julia and Pytest, that can compare functions from Julia and Python to check that the output is between is relatively identical, to each language.

CERTIFICATES

Physiological First AID

Certificate #: 104973770

First AID and CPR/AED Level C

Certificate #: 105003987

TECHNICAL SKILLS

Languages: Python, MATLAB, LaTeX, Julia, HTML, CSS, JavaScript

Libraries: NumPy, Matplotlib, Pandas, SQLAlchemy

Frame Works: Flask, Pytest

Softwares: Onshape, Solidworks, Office360, Google Sheets, Excel, Microsoft VS Code

LANGUAGES

Spanish: Passive bilingualism

EDUCATION

University of Waterloo

Sep. 2024 – April. 2029(Expected)

Bachelor of Science in Mathematical Physics

Waterloo, ON

- Student ID: 21122111

SCHOLARSHIPS AND AWARDS

University of Waterloo President Scholarship

Principal Award - St. Ignatius of Loyola Catholic Secondary Scholar